MILITARY SPECIFICATION

PUMP, FUEL, ELECTRICALLY OPERATED: 24 VOLT DC, IN-TANK TYPE

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

10 SCOPE

l.l Scope. This specification covers electrically operated 24 volt (V), direct current (de) in-tank fuel pumps capable of delivering 200 gallons per hour (gph) with outlet pressure at 4 pounds per square inch gage (psig) (see 6.1).

2* APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

I Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2910
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SPECIFICATIONS
FEDERAL

RR-W- 360	- Wire Fabric, Industrial.
TT-S-735	- Standard Test Fluids; Hydrocarbon.
VV-F-800	- Fuel Oil, Diesel.
CCC-C-421	- Cloth, Denim, Cotton, Shrunk and Unshrunk.

MILITARY

MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-13513	– Terminal, Lug; Splice, Conductor (Electrical, Solderless, for Automotive Use).
MIL-F-13927	- Fungus Resistance Test; Automotive Components.

STANDARDS
FEDERAL

FED-STD-H28	– Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	– Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein- Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

10946915	Pump Assembly.
10947344	Pump Assembly, Fuel (Submersible, 24 V de).

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J962 - Formed Tube Ends for Hose Connections.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2). First article samples shall be inspected by the contractor under the surveillance of the Government to determine conformance to quality assurance provisions of this specification. First article samples shall be fully representative of fuel pumps to be supplied from production tooling and facilities. Any change or deviation of production units from first article sample shall be subject to the approval of the Government.

3.2 Design and construction. In order to assure mechanical and electrical interchangeability, envelope dimensions and construction of the pump shall be within limits shown on figure 1 and Army Drawings 10946915 and 10947344. Location of fuel inlet and outlet ports and electrical connections shall be within limits indicated on figure 1. Pump shall mount vertically in the fuel tank (see 4.3.5).

3.2.1 Materials. Materials shall be as specified herein, in applicable specifications and on applicable drawings (see 6.4).

3.2.2 Standard parts. Military standard parts shall be used whenever possible.

3.2.3 Threaded parts. All screw threads shall conform to FED-STD-H28. The form, number per inch, and class of threads shall be as specified herein, on applicable standards or drawings.

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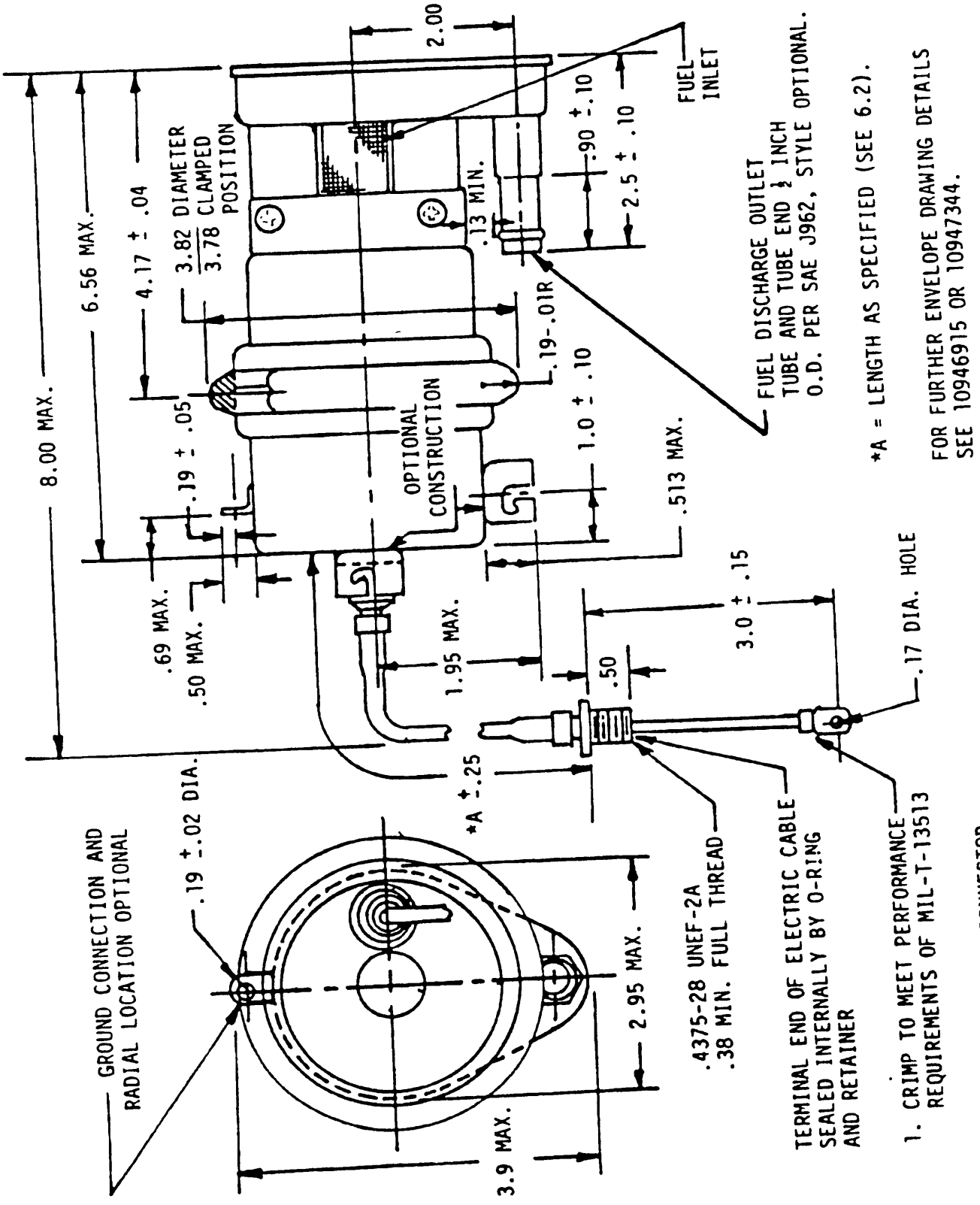


FIGURE 1. Fuel pump envelope.

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3.2.4 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided. Where such contact is not necessary to complete an electrical circuit but is otherwise unavoidable, surfaces shall be insulated as specified for type II, class 3 of MIL-STD-889.

3.2.5 Inlet screen. The inlet area of the pump shall be covered by a screen of 18 X 18 mesh with .017 diameter stainless steel wire in accordance with type I, class optional of RR-W-360.

3.2.6 Cable. The pump shall be provided with a shielded cable, conforming to figure 1, of specified length (see 6.2). The cable shall incorporate a suitable grounding connection for the shielding braid. An independent ground connection shall be used for the pump.

3.3 Performance.

3.3.1 Standard calibration. Fuel pump pressure and flow characteristics shall equal or exceed those illustrated by curves shown in figure 2 (see 4.7.3).

3.3.2 Pumpdown. The pump shall meet pressure and flow rates of figure 2 at all fuel levels 1.5 inches or more above the lowest point of the pump. When pump delivery becomes zero, air pressure developed in the fuel line shdll not exceed 0.5 psig (see 4.7.4).

3.3.3 Dry operation. When tested as specified in 4.7.5, the pump shall be capable of withstanding operation in a dry fuel tank for a period of not less than 4 hours without-evidence of damage or impairment of performance. Subsequent to the period of dry operation, the pump shall meet requirements of 3.3.1*

3.3.4 Electromagnetic compatibility. The fuel pump shall meet radio interference suppression requirements of MIL-STD-461, part 4 pertaining to military vehicles. A radio-frequency noise filter shall be an integral part of the pump (see 4.7.6).

3.3.5 Dielectric strength. When tested as specified in 4.7.7, with the internal ground and radio suppressor ground strap disconnected, the insulation shall show no loosening, cracking, charring, burning, smoking, rupture or other damage.

3.3.6 Vapor separation. When tested as specified in 4.7.8, the pump shall maintain performance within pressure and flow rate limits of figure 2 during the heating cycle. When the pump inlet is removed from the heated fuel and then resubmerged, the pump shall re-establish performance with pressure anti flow rate limits of figure 2 within a period of 5 seconds.

3.3.7 Flow through. When tested as specified in 4.7.9, with the pump electrically disconnected, minimum flow at the pump outlet shall be not less than 60 gph.

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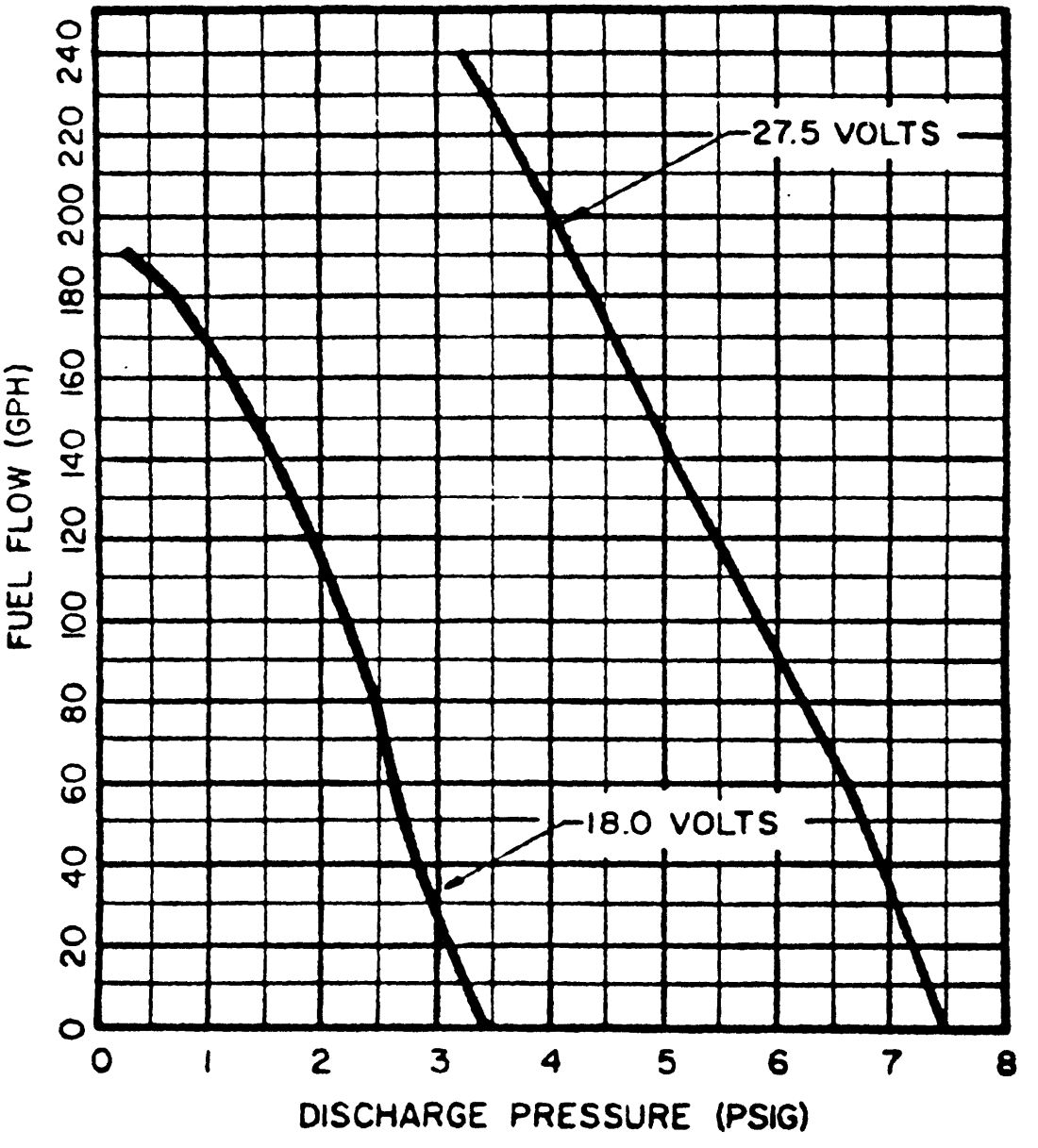


FIGURE 2. Flow and pressure.

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3.3.8 On-off cycling. When tested as specified in 4.7.10, the pump shall meet pressure and flow rates of figure 2 during on cycle portions of the test . A time delay of 3 seconds, after each off cycle, will be allowed to permit pump output to come up to required values.

3.3.9 Endurance. When tested AS specified in 4.7.11, the pump shall meet pressure and flow rates of figure 2. The fuel pump shall require no maintenance during the test. After completion of the endurance test, the pump shall meet requirements of 3.3.1.

3.3.10 Coupling capability (static). When tested as specified in 4.7.12, the pump delivery shall reach a minimum of 200 gph at 4 psig within 60 seconds of restarting.

3.3.10.1 Coupling capability (dynamic). When tested as specified in 4.7,12.1, the fuel pump shall start pumping within 5 seconds after immersion. Decoupling of the impeller shall be considered a failure.

3.3.11 Voltage drop. The millivolt drop between the pump housing and wire braid on the power cable, and between the .4375-28 UNEF-2A thread and wire braid shall meet the requirements of MIL-T-13513 for the wire gage of the center (hot) cable (see figure 1 and 4.7.13).

3.4 Environmental requirements.

3.4.1 Low temperature. When tested as specified in 4.7.14, the fuel pump shall meet pressure and flow rates of figure 2.

3.4.2 High temperature. When tested as specified in 4.7.15, the fuel pump shall meet pressure and flow rates of figure 2.

3.4.3 Explosion-proofness. When tested as specified in 4.7.16, the pump shall not cause ignition of an explosive fuel mixture in the test chamber.

3.4.4 Shock resistance. When tested as specified in 4.7.17, the pump shall withstand, without damage, a shock of 25 gravity units (g) applied once in the direction of each of the 3 major axes. The pump shall subsequently meet requirements of 3.3.1.

3.4.5 Vibration resistance. When tested as specified in 4.7.18, the pump shall evidence no malfunction and shall meet pressure and flow rates of figure ?l

3.4.6 Fungus resistance. Subsequent to testing as specified 4.7.19, the pump shall meet requirements of 3.3.1.

3.4.7 Fuel resistance. Pumps with rubber parts that contact fuel shall withstand, without damage, testing as specified in 4.7.20 and shall . subsequently meet requirements of 3.3.1.

3.4.8 Contaminated fuel. After testing as specified in 4.7.21, the pump shall meet requirements of 3.3.1.

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3.4.9 Corrosion resistance. After testing as specified in 4.7.22, the pump shall meet requirements of 3,3.1.

3.5 Marking. The fuel pump shall be marked in accordance with MIL-STD-130. Marking shall include the following information (see 4.7.2):

a. Operating voltage.
b. Military part number.
c. Name of manufacturer.
d. Date of manufacture.

3.6 Workmanship. Workmanship shaLl be of a quality that will assure a product free of burrs, rust, scratches, chips, sharp edges, loose or defective connectors, cracked insulation, faulty soldering, or other defects which may adversely affect the serviceability or appearance of the pump (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring thal all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent (%) of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

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4.2 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
   1. Examination (see 4.5.2).
   2* Tests (see 4.5.3).
c* Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all examinations and tests shall be performed under conditions specified in 4.3.1 through 4.3.6.

4.3.1 Test fuel. Fuel used for testing shall conform to grade DF-2 of VV-F-800.

4.3.2 Operating voltage. Operating voltage of the fuel pump shall be 27.5 ± 0.1 V dc.

4.3.3 Atmospheric limits. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 25 ± 6°C (77 ± 10°F)
b. Barometric pressure 725 + 50 - 75 millimeter (mm) mercury (HG) (28.5 + 2 - 3 inches)
c. Relative humidity 50 ± 30 percent

4.3.4 Apparatus. Inspection apparatus shall include a fuel tank or tanks suitably equipped to mount the fuel pump within the test fluid at least 2 inches below fluid level. Means shall be provided within the apparatus to recirculate test fuel and to control test fuel temperature at any point between minus 65°F and plus 160°F. The apparatus shall include an electrical source, ranging from 0 to 32 V dc, to operate the fuel pump. The electrical source shall maintain any voltage setting and shall not drop more than 0.5 volts in 1 millisecond (measured with an oscilloscope) when the fuel pump is energized. Monitoring devices shall also be included to determine gph flow rate, psig of pressure generated by the fuel pump, and electrical loads. The pressure gage tap shall be located in a 0.5 inch inside diameter discharge line, 6 to 8 inches from the discharge fitting on the fuel pump. The apparatus shall permit measurement within 10% of product or test specification limits with gage and meter accuracy as specified in 4.3.4.1 and 4.3.4.2, unless closer tolerances are specified herein or in applicable drawings, specifications and standards.

4.3.4.1 Gage and meter accuracy, Gages and meters shall be accurate within the following tolerances:

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Gage or meter	Accuracy tolerance (persentage of scale readings)
Ammeter	within 0.5%
Fl owmeter	within 2.0%
Pressure gage 1/	within 2.0%
Thermometer	within 0.5%
Vol tmeter	within .25% (between 25 and 32 V dc)

1/ Pressure gage reading shall be O with pump stopped and fuel lines full.

4.3.4.2 Gage and meter calibration. Calibration of test equipment shall be conducted at intervals sufficient to maintain specified accuracy. Records of calibrat ion shall be made available to the Government. The Government may refuse to witness an inspection where accuracy of gages and meters is not satisfactorily supported by calibration records (see 4.1.2).

4.3.5 Mounting position. All tests shall be conducted with fuel pump in normal mounting position. Normal mounting position is defined as that position in which the longitudinal axis of the fuel pump is vertical.

4.3.6 Electrical current load. Electrical current load of the fuel pump shall not exceed 4.5 amperes under any condition of operation.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall randomly select 5 fuel pumps properly marked with identifying information from the first 20 pumps produced under the production contract (see 6.2), for first article inspection. First article samples shall be inspected as specified in table I, in the order listed in table II, Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply fuel pumps that are fully representative' of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

TABLE I> Classification of inspections.

Title	Requirement	Inspec-tion	First article	Quality conformance		Con-trol
				Exami-nation	Tests	
Materials and construction	3.2, 3.2.1, 3.2.2, 3.2.3, and 3.2.4	4.7.1	x			
Defects (4.5.2 and table III)	3.2, 3.5 and 3.6	4.7.2	x	x		
Standard calibration	3.3.1	4.7.3	x		x	x
Pumpdown	3.3.2	4.7.4	x		x	
Dry operation	3.3.3	4.7*5	x			x
Electromagnetic compatibility	3.3.4	4.7.6	x			

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TABLE I. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Dielectric strength	3.3.5	4.7.7	X			
Vapor separation	3.3.6	4.7.8	X			
Flow through	3.3.7	4.7.9	X			
On-off cycling	3.3.8	4.7.10	X			X
Endurance	3.3.9	4.7.11	X			
Coupling capability						
(static)	3.3.10	4.7.12	X		X	
(dynamic)	3.3.10.1	4.7.12.1	X		X	
Voltage drop	3.3.11	4.7.13	X			
Low temperature	3.4.1	4.7.14	X			
High temperature	3.4.2	4.7.15	X			
Explosion-proofness	3.4.3	4.7.16	X			
Shock resistance	3.4.4	4.7.17	X			
Vibration resistance	3.4.5	4.7.18	X			X
Fungus resistance	3.4.6	4.7.19	X			
Fuel resistance	3.4.7	4.7.20	X			
Contaminated fuel	3.4.8	4.7.21	X			
Corrosion resistance	3.4.9	4.7.22	X			

TABLE II. Order of first article testing.

Sample number	Paragraph number	Test
1	4.7.12	Coupling capability (static)
	4.7.3	Standard calibration
	4.7.9	Flow through
	4.7.3	Standard calibration
	4.7.5	Dry operation
	4.7.3	Standard calibration
	4.7.10	On-off cycling
	4.7.3	Standard calibration
	4.7.18	Vibration resistance
	4.7.3	Standard calibration
	4.7.17	Shock resistance
	4.7.3	Standard calibration
	4.7.20 1/	Fuel resistance
	4.7.3	Standard calibration

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TABLE II. Order of first article testing - Continued.

Sample number	Paragraph number	Test
2	4.7.12.1	Coupling capability (dynamic)
	4.7.3	Standard calibration
	4.7.4	Pumpdown
	4.7.14	Low temperature
	4.7.15	High temperature
	4.7.3	Standard calibration
	4.7.19	Fungus resistance
	4.7.3	Standard calibration
3	4.7.12	Coupling capability (static)
	4.7.3	Standard calibration
	4.7.11	Endurance
	4.7.3	Standard calibration
	4.7.13	Voltage drop
4	4.7.12.1	Coupling capability (dynamic)
	4.7.3	Standard calibration
	4.7.16	Explosion-proofness
	4.7.3	Standard calibration
	4.7.8	Vapor separation
	4.7.21	Contaminated fuel
	4.7.22	Corrosion resistance
5	4.7.12	Coupling capability (static)
	4.7.3	Standard calibration
	4.7.6	Electromagnetic compatibility
	4.7.7	Dielectric strength

1/ Applicable only to pumps utilizing rubber components in contact with fuel.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

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4.5 Quality conformance inspections.

4.5. \ Sampling.

4.5. 1.} Lot formation. Unless otherwise specified (see 6.2), an inspection lot shall consist of all fuel pumps of one type and part number, from an identitiable production period, from one manufacturer, submitted at one time.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective,

Classification	W
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. for examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.2).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.2).	SIE 1/
103	Screen missing or not as specified (see 3.2.5).	Visual
104	Identification marking, improper (see 3.5).	Visual
105	Faulty workmanship affecting performance (see 3.6).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.2).	SIE 1/
202	Improper marking (see 3.5).	Visual
203	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

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4.5.3 Tests (100%). Each pump shall be subjected to tests specified in table I, in order listed in table IV.

TABLE IV. Order of quality conformance testing.

Paragraph	Test
4.7.3	Standard calibration
4.7.4	Pumpdown
4.7.3	Standard calibration
4.7.12	Coupling capability (static)
4.7.12.1	Coupling capability (dynamic)

4.5.4 Failure. Failure of a quality conformance sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proven to the satisfaction of the Government that any deficiency revealed by examination or test has been corrected.

4.6 Control tests.

4.6.1 Sampling Whe procurement involves less than 100 pumps, 1 of the first 10 pumps produced under contract shall be selected. When procurement involves 100 pumps or more, 1 of first 10 and thereafer 1 of each 500 pumps produced shall be selected, except that not more than 1 pump shall be tested in any given 30 day period.

4.6.2 Test procedure. Each sample selected in accordance with 4.6.1 shall be subjected to tests specified in table 1 in the order listed in table V.

TABLE V. Order of control testing

Paragraph	Test
4.7.5	Dry operation
4.7.3	Standard calibration
4.7.10	On-off cycling
4.7.3	Standard calibration
4.7.18	Vibration resistance
4.7.3	Standard calibration

4.6.3 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proven to the satisfaction of the Government that any deficiency revealed by examination or test has been corrected.

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4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2, 3.2.1, 3.2.2, 3.2.7, and 3.2.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.

4.7.2 Defects. Conformance to 3.2 through 3.6 shall be determined by examination for the defects listed in table III. Examination shall he visual, tactile, or by measurement with standard inspection equipment

4.7.3 Standard calibration. To determine conformance to 3.3.1, the fuel pump shall be mounted in the apparatus, test fuel added, and all fluid and electrical lines connected. The pump shall be operated at 18 and at 27.5 V dc. A record of pressure readings at various flow rates shall be made and compared with performance curves in figure 2.

4.7.4 Pumpdown. To determine conformance to 3.3.2, the fuel pump shall be mounted and operated as in 4.7.3, with the following exceptions:

a. Test fuel discharged by the pump shall not be allowed to re-enter the fuel tank.
b. The pump shall be operated until performance below that specified In figure 2 recurs consistently.
c. The fuel level shall be observed and the level at which pump delivery and discharge pressure deteriorate shall be noted.
d l After pump delivery becomes zero, the discharge line shall be closed and air pressure developed by the pump shall be noted.

4.7.5 Dry operation. To determine conformance to 3.3.3, the fuel pump shall be mounted in the apparatus, test fuel added and all fluid and electrical lines connected. The pump shall be operated at a minimum obtainable discharge pressure. While the pump is operating, the fuel tank shall be drained to a level below the pump inlet. The pump shall be continuously operated without fuel for a period of not less than 4 hours. Thereafter, the fluid level shall be raised until the pump is again submerged 2 inches and the pump shall be tested as specified in 4.7.3.

4.7.6 Electromagnetic compatibility. To determine conformance to 3.3.4, the fuel pump shall be tested for radio interference suppression in accordance with MIL-STD-462.

4.7.7 Dielectric strength. To determine conformance to 3.3.5, the fuel pump motor housing shall be disassembled or cut open to the extent necessary to permit the internal ground connection to be disconnected or isolated and to permit observation of internal wiring. The motor shall be observed while being subjected to 500 volts root mean square (rms) at a frequency of 60 cycles per second (Hz) applied between conductor and ground for a period of 1 minute.

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4 .7.8 Vapor separation. To determine conformance to 3.3.6, the apparatus shall be modified and the fuel pump mounted in accordance with 4.7.8.1. The pump shall be operated in accordance with 4.7.8.2 and performance characteristics noted and compared with figure 2.

4.7.8.1 Vapor separation apparatus The apparatus in 4.3.4 shall be modified by adding means to return vaporized fuel to the tank through condensation. Means shall also be provided to lift or tilt the pump to allow the pump inlet to be raised above the fuel level. The tank shall be filled with fresh fuel (not previously used in high temperature testing) conforming to type II of MIL-G-3056.

4.7.8.2 Procedure. The fuel pump shall be mounted in the tank, all fluid and electrical lines connected and test fuel added (see 4.7.8.1). The fuel shall be heated to 155°F and during this heating period the pump shall be operated so as to maintain delivery at 150 gph. After the system reaches the specified temperature, the pump shall be lifted or tilted to the extent necessary to uncover the inlet area. The pump shall be resubmerged and the time necessary to re-establish flow rate shall be recorded. The pump shall be tilted or lifted and time recorded a total of 5 times.

4.7.9 Flow-through. To determine conformance to 3.3.7, the fuel pump shall be mounted in the tank, all fluid lines connected and test fuel added, Electrical lines shall not be connected. While the pump is electrically and mechanically inert a pressure of 1 psig shall be exerted on the fuel at the pump inlet or, where a pump by-pass system is used, a vacuum of approximately 2.045 inches of mercury shall be applied at the pump outlet. The flow rate developed at the pump outlet shall be noted.

4.7.10 On-off cycling. To determine conformance to 3.3.8, the fuel pump shall be mounted in the tank, all fluid and electrical lines connected and test fuel added. The pump shall be subjected to 1440 cycles of operation. Each cycle shall consist of 15 seconds of operating time followed by 15 seconds of nonoperating time. The pump performance shall be noted during operation for comparison with figure 2.

4.7.11 Endurance. To determine conformance to 3.3.9, the fuel pump shall be mounted in the tank, all fluid and electrical lines connected and test fuel added. The pump shall he operated for a total of not less than 1500 hours in accordance with the schedule set forth in table VI. The pump performance shall be noted for comparison with figure 2. Following completion of the endurance test, the pump shall be tested as specified in 4.7.3.

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TABLE VI. Endurance test schedule.

Fuel grade 1/	Temperature	Test period (hours)
Grade DF-2	77 ± 10°F	1450
Grade DF-A 2/	-60 ± 5°F	50

1/ Specified in VV-F-800.
2/ Same as 1/ except that the cloud point shall be at least -65°F.

4.7.12 Coupling capability (static). To determine conformance to 3.3.10, the pump shall be mounted in the test tank and the tank filled with test fluid to a level of at least 2 inches above the fuel pump. Apply a terminal voltage of 27.5 V dc and adjust pump discharge pressure to 4 psig. Remove terminal voltage (27.5 V dc) and allow pump delivery to return to zero gph. Within a maximum of 30 seconds after' pump delivery returns 'to zero, apply a terminal voltage of 32 V dc.

4.7.12.1 Coupling capability (dynamic). To determine conformance to 3.3.10.1, the basic pump, without brackets and with a 12 inch (min.) discharge hose attached, shall be installed in the tank. The tank shall be filled with test fuel to a level at least 4 inches above the base of the pump . The pump shall be adjusted to deliver 150 gph at 27.5 V dc. The pump shall be lifted at least 3 inches above the fuel. When the pump stops pumping, it shall be forcefully submerged in the fuel to approximately the same depth. The time for the pump to start pumping fuel shall be recorded. This test shall be performed 3 times. In addition, the test shall be repeated 3 times with the vertical axis of the pump at 45° to the fuel surface. All data shall be recorded.

4.7.13 Voltage drop. To determine conformance to 3.3.11, a voltage drop test shall be conducted in accordance with MIL-T-13513.

4.7.14 Low temperature resistance. To determine conformance to 3.4.1, the test tank shall be filled to a level at least 2 inches above the pump. The test fuel shall conform to Grade DF-A of VV-F-800, except that the cloud point shall be -65°F or lower. The temperature of the system shall be lowered to -60 ± 5°F and maintained at that temperature for at least 16 hours. The pump shall then be operated at 18 and 27.5 + 0.1 V dc.

4.7.15 High temperature resistance. To determine conformance to 3.4.2, the test tank shall be filled to a level at least 2 inches above the pump. The system shall be heated to a temperature of 125 + 5°F and maintained at that temperature for at least 4 hours. The pump shall then be operated at 18 and 27.5 + 0.1 V dc.

4.7.16 Explosion-proofness.

4.7. 16.1 Explosion test chamber. The test shall be conducted in a test chamber having the following features:

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a. A pressure release system which operates when an explosion occurs within the chamber.
b. A hinged or removable pressure-sealed door large enough to permit installation of the pump.
c. A means of circulating vapors within the chamber.
d. An observation window.
e. Cable ports with pressure-tight seals.

4.7.16.2 Test procedure. To determine conformance to 3.4.3, an unused (dry) fuel pump shall be installed in the test chamber and gasoline conforming to type I of MIL-G-3056 shall be vaporized and drawn into the chamber. The quantity of gasoline used shall be calculated to result in an air to fuel ratio within limits of 13:1 to 15:1 (by weight). The fuel pump shall be operated for a period of 15 minutes. Immediately after test, the mixture in the chamber shall be ignited to verify that an explosive mixture existed during test period. Withdrawal of a mixture sample for test in a bomb shelter is an allowable procedure.

4.7.17 Shock resistance. To determine conformance to 3.4.4, the fuel pump shall be subjected to method 516, procedure I of MIL-STD-810. A half-sine shock pulse of 25 g shall be applied, for 18 milliseconds nominal duration, once in the direction of each of the 3 major axes. Subsequently, the fuel pump shall be tested as specified In 4.7.3.

4.7.18 Vibration resistance. To determine conformance to 3.4.5, the fuel pump shall be installed in a test tank and the entire system subjected to the vibration test specified in method 201 of MIL-STD-202. If resonance is encountered during this test, the pump shall be vibrated at resonant frequency for a period of 4 hours along each of the 3 major axes. During the test, the pump shall meet pressure and flow rates of figure 2.

4.7.19 Fungus resistance. To determine conformance to 3.4.6, the fuel pump shall be subjected to class 1, method B fungus resistance test of MTL-F-13927, except that incubation shall be for a single period of 90 days. Subsequently, the pump shall be tested as specified in 4.7.3.

4.7.20 Fuel resistance. To determine conformance to 3.4.7, any pump utilizing rubber parts that come in contact with fuel shall be subjected to the test specified in table VII. Except for the first 48 hours of phase 1 and phase III testing, the pump shall be aerated at rated flow for 2 hours per day. Subsequently, the pump shall be tested as specified in 4.7.3.

TABLE VII. Fuel resistance schedule.

Phase	Hours	Medium	Temperature
I Soak	168	TT-S-735, Type III	75 + 5°F
II Dry	4	Air	155 + 5°F
111 Soak	168	TT-S-735, Type III	75 + 5°F
IV Dry	4	Air	155 + 5°F

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4.7.21 Contaminated fuel. To determine conformance to 3.4.8, the pump shall be mounted in a fuel tank and the tank filled with contaminated test fluid of type and concentration specified in table VIII and in accordance with TT-S-735. The tank shall be filled to a level at least 2 inches above the fuel pump. The pump shall be operated to deliver a minimum of 220 gph for a period of 3 hours and subsequently 25 gph for another period of 3 hours. The pump shall then be flushed out with clear test fluid and tested in accordance with 4.7.3.

TABLE VIII. Contaminated fuel endurance test dust.

Contaminant	Particle size	Quantity
Iron oxide	0-5 microns	28.5 gm/400 gal.
Iron oxide	5-10 microns	1.5 gm/400 gal.
Sharp silica sand	150-300 microns	1.0 gm/400 gal.
Sharp silica sand	300-420 microns	1.0 gm/400 gal.
Prepared dirt conforming to AC Spark Plug Co. Part No. 1543637 (coarse Arizona road dust) or equivalent (see 6.4).	Mixture as follows: 0-5 microns (12%) 5-10 microns (12%) 10-20 microns (14%) 20-40 microns (23%) 40-80 microns (30%) 80-200 microns (9%)	8.0 gm/400 gal.
Lint 1/	As collected	0.1 gm/400 gal.
Crude napthenic acid		0.03% by volume
Salt water (solution shall contain 4 parts NaCl to 96 parts H_2O by weight).		0.01% entrained

1/ Obtained by tumbling laundered cloth, CCC-C421, type I, class B-S, in a tumbling machine (clothes drier).

4.7.22 Corrosion resistance. To determine conformance to 3.4.9, the pump shall be subjected to method 101 of MIL-STD-202 for a period of 100 hours, except that a 20% salt solution shall be used.

4.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of paragraph 5.1 of this specification.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

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6. NOTES

6.1 Intended use. The fuel pump procured by this specification is intended for use in the fuel systems of military vehicles.

6.2 ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
h. If first article samples are not required (see 3.1, 4.4 and 6.3).
c. Length of cable required (see 3.2.6).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If atmospheric limits shall be other than as specified (see 4.3.3).
g. If number of first article samples shall be other than as specified (see 4.4).
h. If number of lot samples shall be other than as specified (see 4.5.1.1).
i. Selection of applicable level of preservation, packaging and packing (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2.1).

6.4 Contaminated solid. The AC Spark Plug Division, Coarse Air Cleaner Dust , Package No. 1543637 has been satisfactory for fuel pump testing (see table VIII).

6.5 Subject term (key word) listing.

Electrically operated fuel pump: 24 Volt DC, in-tank type
Fuel pump, electrically operated: 24 Volt DC, in-tank type
In-tank fuel pump, electrically operated: 24 Volt DC

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due LU the extensiveness of the changes.

Custodian:
Army - AT

Review activity:
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